



Liver Function Tests

Biochemistry GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which two enzymes are most liver-specific (ALT) vs. also found in heart/muscle/kidney (AST)?
2. What's the difference between what ALP and GGT each tell you when both are raised together vs. GGT alone?
3. Which liver-made clotting factors does PT/INR depend on, and what vitamin do they need?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Why test the liver?

Liver in brief	Largest gland; upper right abdomen; ~1.5 kg; performs 500+ functions — digestion (bile), metabolism, detoxification, storage, synthesis, immune function, hormone regulation, waste removal, energy regulation
Liver function tests (LFTs)	A blood panel measuring enzymes, proteins & other substances to assess liver health/function — used to diagnose liver disease, monitor severity, and check for damage

2 Liver enzymes — the core discriminator

All four rise with liver injury, but each has a different specificity — that's the exam skill.

Enzyme	Normal range	Source	Raised in
ALT	5–45 IU/L	Mainly liver — most liver-specific	Viral hepatitis, ischaemic hepatitis, drug/chemical injury
AST	5–45 IU/L	Liver, heart, kidney, skeletal muscle, brain — less liver-specific	Hepatitis, cirrhosis, MI, muscular dystrophy, kidney disease
ALP	30–120 IU/L	Liver, bone, bile ducts, placenta — non-specific	Liver inflammation, bile duct blockage, bone disease
GGT	5–50 IU/L	Liver, kidney, biliary tract; used for glutathione synthesis	Liver & bile duct disorders, chronic alcohol use

AST/ALT ratio pearl

- **AST > ALT** → alcoholic liver disease
- **ALT > AST** → viral or non-alcoholic liver disease

3 Total protein — albumin & globulin

	Normal range	Made by	Changes
Albumin	3.5–5 g/dL	Most abundant protein synthesised by the liver	↓ in all chronic liver disease (cirrhosis, hepatitis, liver cancer); high levels rarely due to liver problems
Globulin	2.5–3.5 g/dL	Liver + immune system (γ-globulins/immunoglobulins, α & β globulins, clotting globulins)	↑ γ-globulins in chronic liver disease (hepatitis, cirrhosis), infection, multiple myeloma

4 Bilirubin

What it is	Byproduct of RBC breakdown; the pigment behind jaundice
Total bilirubin	0.2–1.2 mg/dL
Unconjugated (indirect)	0.2–0.8 mg/dL — bound to albumin

Conjugated (direct)	0.0–0.4 mg/dL
High levels	Jaundice — mainly from gallstones, or acute/chronic hepatitis

5 Prothrombin time (PT) / INR

What it tests	The extrinsic coagulation pathway — clotting factors II, V, VII, X, all liver-made and vitamin K-dependent
PT	Time for blood to clot, in seconds — normal 11–13.5 sec
INR	Standardised PT, comparable across labs — normal ≈ 1.0
Prolonged in	Liver disease (hepatitis, cirrhosis, acute liver failure), vitamin K deficiency, clotting factor deficiency, warfarin therapy

Practical measurement: prepare hands/device → match the test-strip code → prepare the finger → finger-stick → apply blood to the strip → wait for and review the result.

6 What LFTs are used for

Diagnosis & monitoring	Diagnose and monitor liver disease severity
Drug monitoring	Evaluate liver function in patients on medications that can affect the liver
Symptom work-up	Investigate jaundice, abdominal pain, or unexplained fatigue
Toxin/alcohol screen	Check for liver damage from toxins or excess alcohol

7 Worked cases — pattern recognition

CASE 1 — OBSTRUCTIVE PATTERN

58-year-old man: jaundice, pruritus, pale stools, dark urine. Ultrasound: gallstone in the common bile duct.

AST 98 IU/L · **ALT** 110 IU/L · **ALP** 850 IU/L · **Total bilirubin** 13 mg/dL (mainly conjugated) · **Albumin** 4.1 g/dL · **PT/INR** 1.3

Interpretation: ALP and bilirubin are very high, AST/ALT only mildly raised → **obstructive (cholestatic) jaundice**, consistent with the gallstone blocking the common bile duct.

CASE 2 — HEPATOCELLULAR / FULMINANT PATTERN

30-year-old woman: confusion, jaundice, history of paracetamol overdose.

AST 2300 IU/L · **ALT** 2550 IU/L · **ALP** 135 IU/L · **Total bilirubin** 8.5 mg/dL (mainly conjugated) · **Albumin** 3.7 g/dL · **PT/INR** 4

Interpretation: Massively raised AST/ALT with markedly prolonged INR and encephalopathy (confusion) → **acute fulminant hepatic failure**.

Discriminator: **ALP** $\gg$ **AST/ALT** points to an obstructive/cholestatic pattern; **AST/ALT** $\gg$ **ALP**, especially with a prolonged INR, points to a hepatocellular/fulminant pattern.

P Must-remember pearls

- **ALT** is more liver-specific; **AST** is also found in heart, kidney, muscle, brain. AST > ALT suggests alcoholic liver disease; ALT > AST suggests viral/non-alcoholic disease.
- **ALP** (liver, bone, bile duct, placenta) rising far more than AST/ALT signals an **obstructive/cholestatic** pattern — as in gallstones blocking the common bile duct.
- **Albumin** falls in chronic liver disease; it's the most abundant liver-synthesised protein.
- PT/INR reflects the **extrinsic pathway** (factors II, V, VII, X) — all liver-made and vitamin K-dependent; a very prolonged INR with sky-high AST/ALT suggests fulminant hepatic failure.
- Bilirubin is the pigment of jaundice; total = unconjugated (indirect, albumin-bound) + conjugated (direct).

Q Test yourself

1. Which enzyme is the most liver-specific marker of hepatocyte injury?

- A) AST
- B) ALT
- C) ALP
- D) GGT

2. An AST:ALT ratio greater than 1 (AST higher than ALT) is classically associated with:

- A) Viral hepatitis
- B) Alcoholic liver disease
- C) Bone disease
- D) Gallstone obstruction

3. A patient with a gallstone obstructing the common bile duct would most likely show:

- A) Markedly raised ALP and bilirubin with only mild AST/ALT rise
- B) AST/ALT in the thousands with normal ALP
- C) Low albumin with normal bilirubin
- D) Prolonged INR with normal bilirubin

4. PT/INR primarily evaluates the function of clotting factors:

- A) VIII, IX, XI, XII
- B) II, V, VII, X
- C) I, XIII
- D) Von Willebrand factor only

5. A patient with a paracetamol overdose develops AST/ALT in the thousands and a markedly prolonged INR with confusion. This picture suggests:

- A) Gallstone obstruction
- B) Acute fulminant hepatic failure
- C) Chronic stable cirrhosis
- D) Isolated bone disease

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. **B — ALT** — found mainly in the liver, unlike AST which is also in heart/muscle/kidney/brain.
- 2. **B — Alcoholic liver disease** — AST > ALT is the classic alcoholic pattern.
- 3. **A — High ALP/bilirubin, mild AST/ALT** — the obstructive/cholestatic pattern, as in Case 1.
- 4. **B — II, V, VII, X** — the extrinsic pathway factors, all liver-synthesised and vitamin K-dependent.
- 5. **B — Acute fulminant hepatic failure** — massive transaminase rise + coagulopathy + encephalopathy, as in Case 2.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Biochemistry GIT Block · MEDucation by Mattin Majid · Liver Function Test, Dr Fargeen E. ALmukhtar
Study alongside the lecture, not instead of it.